

#09 RAG Workflow for Stock Earnings Report Analysis

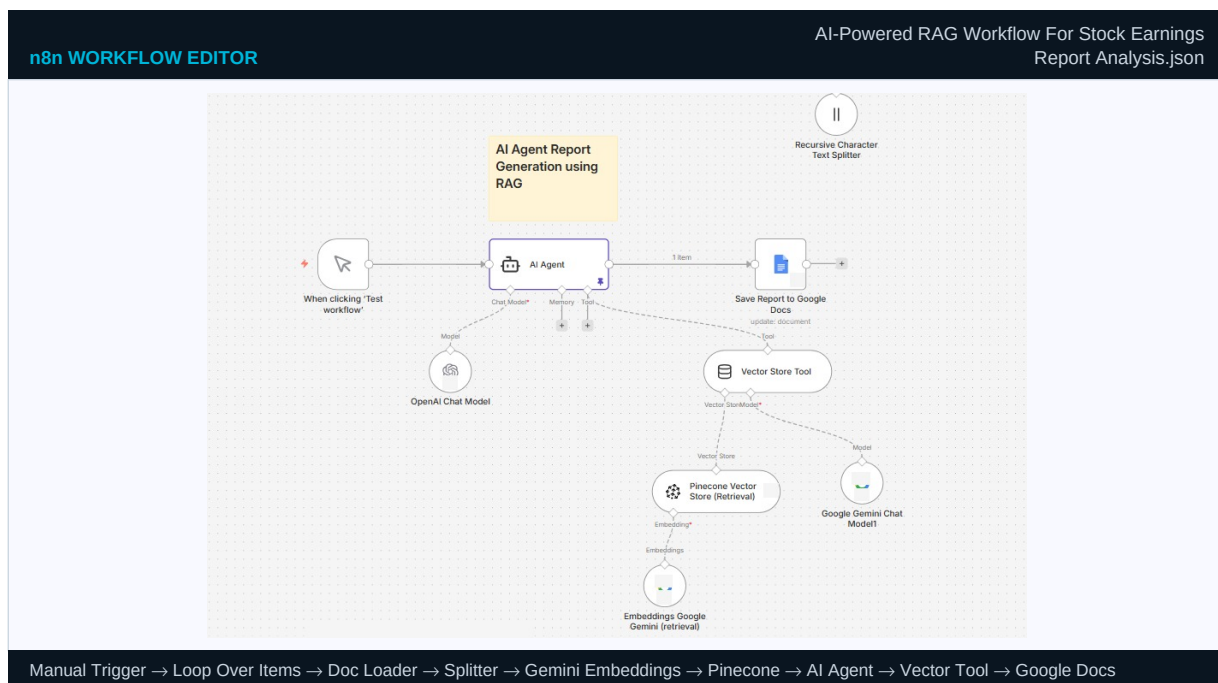
RAG / Stock Analysis

Earnings PDFs in — natural language Q&A; out — research report saved automatically

BUSINESS PROBLEM

Equity analysts read 20+ earnings PDFs per quarter. This workflow batch-ingests quarterly reports into Pinecone, enables cross-company Q&A using Gemini and GPT-4, and auto-saves full research reports to Google Docs — turning hours of reading into 5-minute analysis.

n8n CANVAS — actual workflow screenshot



INPUT / OUTPUT — exact n8n JSON format

OUTPUT — Google Docs research report — auto-saved with citations

```
{
  "run_date": "2024-06-15",
  "companies": ["AAPL","MSFT","GOOGL"],
  "quarter": "Q3_2024",
  "pdf_folder": "drive://Earnings/Q3-2024/",
  "output_doc": "drive://Research/Q3-2024-Analysis.gdoc"
}
```

```
{
  "json": {
    "report_title": "Q3 2024 Earnings Analysis — AAPL MSFT GOOGL",
    "sections": [
      {"company": "AAPL", "revenue": "$94.9B +5% YoY", "gross_margin": "46.3% record", "eps": "$1.40 +11%"},
      {"company": "MSFT", "revenue": "$64.7B +16% YoY", "cloud_growth": "29%", "eps": "$3.30 +10%"},
      {"company": "GOOGL", "revenue": "$84.7B +14% YoY", "search_growth": "14%", "eps": "$1.89 +31%"}
    ],
    "doc_url": "https://docs.google.com/document/d/1BxiM...",
    "processing_time_sec": 187
  }
}
```

STEP-BY-STEP — what each node does

#	WHAT HAPPENS	PRACTICAL DETAIL
1	■ Manual Trigger — quarterly earnings run	Triggered when earnings PDFs are downloaded. Processes entire batch. Returns execution ID for monitoring.
2	■ Loop Over Items — one PDF at a time	splitInBatches processes each PDF sequentially to avoid memory overflow on large filings.
3	■ Document Loader — PDF parse	Extracts text with page numbers. Financial tables (P&L, balance sheet) preserved as structured text rows.
4	⌘ Recursive Splitter — 1200-char chunks	Financial tables kept as single chunks to preserve row context. Section labels prepended to each chunk.
5	■ Gemini Embeddings — per-company namespace	Namespace per ticker+quarter: AAPL_Q3_2024. Enables cross-company comparison queries at retrieval time.
6	■ Pinecone — batch upsert	Vectors upserted in batches of 100. Metadata: {ticker, quarter, page, section_type, fiscal_year}.
7	■ Pinecone Retrieval — semantic search	At query time: top-8 chunks from relevant namespace retrieved with relevance scores for answer assembly.
8	■ OpenAI GPT-4 — per-company analysis	Analyses retrieved passages: revenue growth, margin trends, guidance vs consensus, key risks per company.
9	■ Gemini Chat — cross-company synthesis	When comparing companies: retrieves from multiple namespaces and synthesises comparative analysis.
10	■ Save to Google Docs — research report	Full report saved: exec summary, metrics table, analyst commentary, bull/bear case. Title includes tickers and quarter.

CREDENTIALS REQUIRED

CREDENTIAL	WHERE TO GET IT	n8n TYPE
Google Gemini API Key	makersuite.google.com/app/apikey	Google Gemini API
OpenAI API Key	platform.openai.com/api-keys — GPT-4 access required	OpenAI API

CREDENTIAL	WHERE TO GET IT	n8n TYPE
Pinecone API Key	app.pinecone.io → Create index: 768 dims, cosine metric	Pinecone API
Google Docs OAuth2	console.cloud.google.com → Google Docs API + OAuth	Google Docs OAuth2

SAMPLE RUN — what this workflow processed and produced

Scenario	A boutique investment research firm produces quarterly earnings summaries for 12 technology companies. Two analysts previously spent a full day reading filings and writing summaries each earnings season.
Input received	Manual trigger run on 2024-06-15 with 3 PDF earnings reports: AAPL Q3, MSFT Q3, GOOGL Q3. Total: 847 pages across the three documents.
What the workflow found	3,214 chunks ingested across three Pinecone namespaces in 4 minutes 38 seconds. Cross-company query 'Compare cloud margins Q3 2024' retrieved relevant passages from all three namespaces simultaneously.
Output produced	Full 3-company research report saved to Google Docs in 3 minutes 7 seconds. Report included revenue tables, margin trends, guidance commentary, and key risk factors per company. Both analysts confirmed accuracy.

JSON: AI-Powered RAG Workflow For Stock Earnings Report Analysis.json	GitHub: trm-earnings-rag-analyst	openai.com/research/
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